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1.Introduction

- **IUCN -** The International Union for the Conservation of Nature (or IUCN) is one of the oldest conservation organizations in the world. The IUCN is responsible for producing the IUCN Red List of Threatened Species, which was developed more than 50 years ago, and is currently the global gold standard for assessing the impacts of threats on species global and regional populations.
- **Marine Mammals Taxonomy -** There are 187 different species of marine mammals on our planet. In biology, each unique species (or organism) is known by their two-word Latin names, which are comprised of the name of the genus in which they are found combined with a second name called the species name. Here, this Orca Whale has a Latin name Orcinus orca, which is the genus name Orcinus combined with the species name orca.

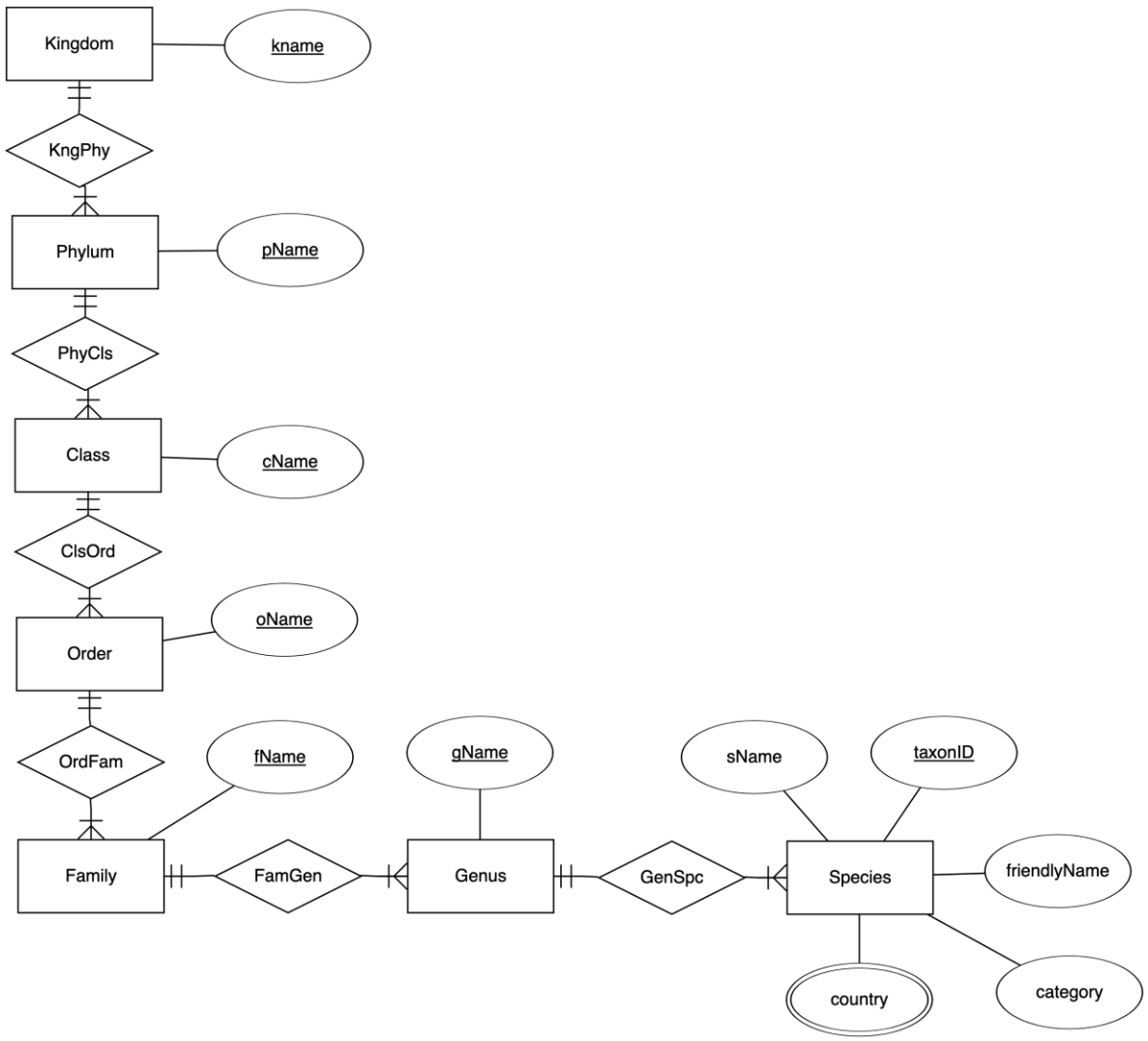
  Kingdom - Animalia
  Phylum – Chordata
  Class – Mammalia
  Order – Cetacea
  Family – Delphinidae
  Genus – Orcinus
  Species – orca

- **IUCN standardized scientific criteria for the species relative risk of extinction -** the IUCN Red List, species are assessed by groups of scientists under standardized, scientific Criteria for their relative risk of extinction, expressed as a Red List Category. This category is ranging from Data Deficient to Extinct. A species is considered threatened when it is classified as either Vulnerable, Endangered or Critically Endangered. [CR, EN, VU]. Other categories can include Near Threatened [NT], which means it comes close to qualifying for a threatened category but doesn't fulfill all the Criteria, or Least Concern [LC] meaning it does not meet any criteria for being threatened, or Data

Deficient [DD], which means we don't have enough information to know if it is threatened or not.

- Species Country Occurrences

2. ER Diagram



3. Requirements Description

The IUCN Red list of threatened species database stores information regarding the taxonomy of marine mammal global species, red list category and country occurrences.

- **Taxonomy** is the scientific study of naming, defining and classifying groups of biological organisms based on shared characteristic. The taxonomy of species contains information about Kingdom, Phylum, Class, Order, Family, Genus and species.
- **kingdom** is the second highest taxonomic rank. In this database it is uniquely identified by kingdom name. Kingdoms are divided into smaller groups called phyla.
- **Phylum** is a level of classification or taxonomic rank below kingdom and above class, and it is uniquely identified by phylum name.
- **class** is a taxonomic rank, as well as a taxonomic unit, a taxon, in that rank. It is a group of related taxonomic orders and it is uniquely identified by class name.
- **Order** is one of the eight major hierarchical taxonomic ranks in Linnaean taxonomy and it is uniquely identified by order name.
- **Family** is classified between order and genus. In this database it is uniquely identified by family name.
- **Genus** name forms the first part of the binomial species name for each species within the genus. Genus is uniquely identified by genus name.
- **Species** is the basic unit of classification and a taxonomic rank of an organism, as well as a unit of biodiversity. Here it is uniquely identified by taxonID. Other characteristics includes species name, friendly name, category and country.
- On the basis of the extinction rate, species are categorized into 8 categories, expressed as red list category.

4. ER Diagram Uncaptured Constraints

The following is a list of constraints that are not captured by the ER diagram.

- Not null: all fields are required to have a value.
- Unique: There is no unique key except the primary key.

5. Relational Schema: Syntax Summary and Table Details

Kingdom (kName)

Phylum (pName , kName)
Foreign key (kName) references Kingdom(kName)

Class (cName, pName)
Foreign key (pName) references Phylum(pName)

Order (oName, cName)
Foreign key (cName) references Class(cName)

Family (fName, oName)
Foreign key (oName) references Order (oName)

Genus (gName, fName)
Foreign key (fName) references Family(fName)

Species (taxonID, sName, friendlyName, category, gName)
Foreign key (gName) references Genus(gName)

SpCountry (taxonID, country)
Foreign key (taxonID) references Species(taxonID)

Table Name	Attribute	Description
Kingdom	kName	Name of the Kingdom
Phylum	pName	Name of the Phylum
	kName	Name of the Kingdom
Class	cName	Name of the Class
	pName	Name of the Phylum
Order	oName	Name of the Order
	cName	Name of the Class
Family	fName	Name of the Family
	oName	Name of the Order
Genus	gName	Name of the Genus
	fName	Name of the Family
Species	taxonID	Unique ID of Taxonomy
	sName	Name of the species
	Friendly Name	Latin name of the species
	Category	Red List Catagory
	gName	Name of the Genus
SpCountry	taxonID	Taxonomy ID
	Country	Name of the country